BANKS-SP / BANK-A

BANKS-SP / BANK-B

BANKS-SP / SP-MATRIX

BANKS-SP / BLEEDERS

CONTROL / CARD-IF

CONTROL / COIL-DRIVER

CONTROL / INTERCONNECT

COMMS-HMI / CAN

COMMS-HMI / HMI

LLC-LEGS / BUS-IN

LLC-LEGS / LEG-1

LLC-LEGS / LEG-2

LLC-TANKS / TANK

LLC-TANKS / RECT-A

LLC-TANKS / RECT-B

LLC-TANKS / FT1-WINDOW

OUTPUT-GENSING / OUTPUT

OUTPUT-GENSING / SENSE-VBKA

OUTPUT-GENSING / SENSE-VBKB

OUTPUT-GENSING / SENSE-VOUT

OUTPUT-GENSING / ISO-BIAS

OUTPUT-GENSING / NTC

NET NAMING

Internal junctions are named for what they join

Un=Un_=PWn- node at that IC pin e.g. U3=U3_ VINP

Rxdetect_M midpoint of a series pair e.g. RBAL7A_M

Rxdetect_=cmc- tap between Rxdetectcmcfe e.g. RVTO_S1

all others are explicit design nets

MAGNETICS CONSTRUCTION

Identity per docs/magnetics.md (turns, gap, litz, acceptance lines live there)

D1 T1A/T1B (one D): 2 cells, primaries in series -> 2; each 3x ETD72/32/2 (3-set former), 0.45/4, p/s 0/18/s0.07/ t = sec 2x foil 0.08x28 halves, Lm 17.8uH/set +/-7%, Bpk 150mT, plate/web-BONDED

D2 L/B (cw P): external L= 2.33uH +/-3%, 2x ETD72/3/2, 9t+2, litz 1200/0x0.05, distributed gap, bonded

total Lr = D2 + 2x cell leakage (acceptance +/-10%) + 5.1uH loop + w/ 0% simulated; Cr at TANK

bond loss screened by the EOL bonded thermal scan (T_NTMR NTC rise at fixed load)

SHEET INDEX

1k kW air DC-DC (upper) - 2 of 3

OUTPUT-SENSING - 6 sections

BANKS-SP - 4 sections

LLC-TANKS - 4 sections

CONTROL - 3 sections

LLC-LEGS - 3 sections

COMMS-HMI - 2 sections

rev D.4 - 22 sections - 215 components

DC-Modules 50 kW air — DC-DC board (full-bridge LLC)

DC-Modules 50 kW air - board DC-DC (upper), sheet 2 of 3

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